

Theorem 1.3. *Suppose X admits a weakly reducible genus-three trisection $\mathcal{T}$. Then either $\mathcal{T}$ is reducible, or $\mathcal{T}$ contains a five-chain. In particular, X is diffeomorphic to a spun lens space S_p or its sibling S'_p , S^4 , or a connected sum of copies of $\pm\mathbb{CP}^2$, $S^1 \times S^3$, and $S^2 \times S^2$.*

A trisection $\mathcal{T}$ is determined up to diffeomorphism by its *spine*, the union of three 3-dimensional handlebodies $H_\alpha \cup H_\beta \cup H_\gamma$. A trisection $\mathcal{T}$ is said to be *reducible* if there is a curve c bounding disks in all three handlebodies; in this case, we can split $\mathcal{T}$ into a connected sum of smaller-genus trisections. In a foundational result from Heegaard theory, Casson and Gordon proved that if a 3-manifold Y admits a weakly reducible Heegaard splitting, then either the splitting is reducible or Y contains an essential surface [CG87]. We adapt weak reducibility to the setting of trisections, providing further evidence that key ideas from Heegaard theory can inform new insights in dimension four via trisections. We say that $\mathcal{T}$ is *weakly reducible* if there are disjoint non-separating curves c and c' such that c bounds a disk in one of the three handlebodies, and c' bounds a disk in the other two. See [ST94] for additional discussion of weak reduction in the context of Heegaard splittings.

A *five-chain* is a specific sequence of five curves bounding disks in each of the three handlebodies such that consecutive curves intersect once and other pairs of curves are disjoint. A more rigorous definition can be found in Section 5, in which we show that if $\mathcal{T}$ admits a five-chain, then X can be obtained by surgery on a loop ℓ in a manifold X' admitting a lower-genus trisection $\mathcal{T}'$. The proof of Theorem 1.3 relies heavily on ideas from the theory of Heegaard splittings and uses a variety of tools and techniques, including connections between 4-manifolds and Dehn surgery on knots in 3-manifolds, weak reduction and thin position of Heegaard splittings, and many ideas arising in the proof of Theorem 1.1.

In the spirit of classifying as many genus-three trisections as possible, we prove a stronger version of Theorem 1.3 using the same techniques. Given a trisection $\mathcal{T}$ for X with spine $H_\alpha \cup H_\beta \cup H_\gamma$, a *dependent triple* for $\mathcal{T}$ consists of three pairwise disjoint non-separating curves α_1 , β_1 , and γ_1 bounding disks in H_α , H_β , and H_γ , respectively, such that the homology classes $[\alpha_1]$, $[\beta_1]$, and $[\gamma_1]$ are linearly dependent in $H_1(\Sigma)$.

Theorem 1.4. *Suppose a genus-three trisection $\mathcal{T}$ of X admits a dependent triple. Then either $\mathcal{T}$ is reducible, or $\mathcal{T}$ contains a five-chain. In particular, X is diffeomorphic to a spun lens space S_p or its sibling S'_p , S^4 , or a connected sum of copies of $\pm\mathbb{CP}^2$, $S^1 \times S^3$, and $S^2 \times S^2$.*

Corollary 1.5. *If X is simply-connected and admits a genus-three trisection $\mathcal{T}$ with a dependent triple, then X is diffeomorphic to S^4 or a connected sum of copies of $\pm\mathbb{CP}^2$, and $S^2 \times S^2$.*

Corollary 1.5 is notable in light of the trisection genus additivity problem, which asks whether $g(X_1 \# X_2) = g(X_1) + g(X_2)$. In particular, there are examples of exotic $\mathbb{CP}^2 \#^2 \overline{\mathbb{CP}}^2$ (manifolds X homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \#^2 \overline{\mathbb{CP}}^2$) [AP10, FS11]. If trisection genus is additive, then the fact that there is some k such that $X \#^k (S^2 \times S^2)$ is diffeomorphic to $\mathbb{CP}^2 \#^2 \overline{\mathbb{CP}}^2 \#^k (S^2 \times S^2)$ [Wal64] implies that $g(X) = g(\mathbb{CP}^2 \#^2 \overline{\mathbb{CP}}^2) = 3$. Corollary 1.5 can be interpreted as evidence that no exotic $\mathbb{CP}^2 \#^2 \overline{\mathbb{CP}}^2$ admits a genus-three trisection, which would imply that trisection genus is not additive. For more information on the additivity of trisection genus and exotic pairs, see Section 1.3 of [LCM22].

Remark 1.6. The only progress on the genus-three classification problem appears in work of the first author and Moeller in [AM22]. In Section 7, they prove that a special class of genus-three trisections, called *Farey trisections*, correspond to standard 4-manifolds (and